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1. Area of a Rectangle

Question:Write a Python program that accepts the length and width of a rectangle from the user and

formula : $A = \text{length} \times \text{width}$

```
In [39]: length = int(input('Enter the lenth of rectangle :'))
width = int(input('Enter the width of rectangle :'))

area_of_rectangle = (lenth* width)
area_of_rectangle

print('length of rectangle : {}, width of rectangle : {} area_of_rectangle : {}'.format(length, width, area_of_rectangle))
print(f'length : {length}, width : {breadth} area_of_rectangle : {area_of_recta
```

```
length of rectangle : 10, width of rectangle : 8 area_of_rectangle : 90
length : 10, width : 8 area_of_rectangle : 90
```

2. Perimeter of a Rectangle

Question:Write a Python program that takes the length and width of a rectangle and calculates its perimeter.

formula: $P = 2(l + w)$

```
In [38]: length = int(input('enter the length of rectangle :'))
width = int(input('enter the width of rectangle :'))
Perimeter = 2 * (length + width)

print('length of rectangle : {}, width of rectangle : {}, Perimeter of a rectang
print(f'length : {length}, width : {width}, Perimeter of a rectangle : {perimete

print('length of rectangle is :',length)
print('width of rectangle is :',width)
print('Perimeter of a rectangle :',Perimeter)
```

```
length of rectangle : 10, width of rectangle : 9, Perimeter of a rectangle : 40.0
length : 10, width : 9, Perimeter of a rectangle : 40.0
length of rectangle is : 10
width of rectangle is : 9
Perimeter of a rectangle : 38
```

3. Area of a Square

Question:Write a Python program that reads the side length of a square and finds its area.

formula : $A = side * side$

```
In [37]: side = int(input('Enter the side value of square :'))
area_of_square = side*side

print('side length of a square : {} , area_of_square : {}'.format(side,area_of_s
print(f'side : {side} , area_of_square : {area_of_square}')
```

```
side length of a square : 40 , area_of_square : 1600
side : 40 , area_of_square : 1600
```

4. Perimeter of a Square

Question:Write a Python program that takes the side length of a square and calculates its perimeter.

formula : $P=4s$

```
In [43]: side = float(input("Enter side length: "))

perimeter = 4 * side

print("Perimeter of Square =", perimeter)

print('side length of a square : {} , perimeter of a square : {}'.format(side,pe
print(f'side length : {side} , perimeter of a square : {perimeter}')
```

```
Perimeter of Square = 40.0
side length of a square : 10.0 , perimeter of a square : 40.0
side length : 10.0 , perimeter of a square : 40.0
```

5. Area of a Circle

Question:Write a Python program that accepts the radius of a circle and calculates its area.

formula : $A = 3.14 * r * r$

```
In [45]: r=int(input('Enter the value for r :'))
area_of_circle = 3.14*r*r
area_of_circle

print('radius of a circle :{}, area_of_circle : {}'.format(r,area_of_circle))
print(f'radius:{r}, area_of_circle : {area_of_circle}')
```

```
radius of a circle :30, area_of_circle : 2826.0
radius:30, area_of_circle : 2826.0
```

6. Circumference of a Circle

Question:Write a Python program that takes the radius of a circle and calculates its circumference.

formula: $C=2\pi r$

```
In [46]: radius = float(input('Enter the radius:'))
circumference_of_circle = 2 * 3.14 * radius
circumference_of_circle

print('radius of a circle : {},circumference of circle : {}'.format(radius,circu
print(f'radius : {radius},circumference of circle : {circumference_of_circle}')
```

radius of a circle : 9.0,circumference of circle : 56.52
radius : 9.0,circumference of circle : 56.52

7. Area of a Triangle

Question:Write a Python program that accepts the base and height of a triangle and calculates its area.

formula: $A=1/2bh$

```
In [47]: base= eval(input('enter the base:'))
height= eval(input('enter the height:'))
Area_of_Triangle = 0.5 * base * height
Area_of_Triangle

print('base of a triangle : {},height of a triangle : {},Area_of_Triangle : {}'.f
print(f'base: {base} ,height:{height},Area_of_Triangle : {Area_of_Triangle}')
```

base of a triangle : 10,height of a triangle :9,Area_of_Triangle : 45.0
base: 10 ,height:9,Area_of_Triangle : 45.0

8. Simple Interest

Question:Write a Python program that accepts principal amount, rate of interest, and time period and calculates simple interest.

SI= $P \times R \times T / 100$

```
In [49]: principal_amount = float(input('Enter principal amount :'))
rate_of_interest = float(input('Enter rate of interest :'))
time_period = float(input('Enter time period :'))
simple_interest = (principal_amount * rate_of_interest * time_period) / 100
simple_interest

print('principal_amount : {} , rate_of_interest : {} , time_period : {}, simple_
print(f'principal_amount : {principal_amount} , rate_of_interest : {rate_of_inte

principal_amount : 200000.0 , rate_of_interest : 5.0 , time_period : 2.0, simple_
interest : 20000.0
principal_amount : 200000.0 , rate_of_interest : 5.0 , time_period : 2.0, simple_
interest : 20000.0
```

9. Total Amount Using Simple Interest

Question:Write a Python program that calculates the final amount after adding simple interest to the principal.

formula : $\text{Amount} = P + SI$

```
In [51]: principal_amount = float(input('Enter principal amount :'))
rate_of_interest = float(input('Enter rate of interest :'))
time_period = float(input('Enter time period :'))

simple_interest = (principal_amount * rate_of_interest * time_period) / 100
Total_amount = principal_amount + simple_interest

print("Simple interest =", simple_interest)
print("Total amount =", Total_amount )

print('principal_amount : {} , rate_of_interest : {} , time_period : {} , simple_
print(f'principal_amount : {principal_amount} , rate_of_interest : {rate_of_inte

Simple interest = 680.0
Total amount = 34680.0
principal_amount : 34000.0 , rate_of_interest : 2.0 , time_period : 1.0, simple_i
nterest : 680.0 ,Total_amount : 34680.0
principal_amount : 34000.0 , rate_of_interest : 2.0 , time_period : 1.0, simple_i
nterest : 680.0,Total_amount : 34680.0
```

10. Average of Three Numbers

Question:Write a Python program that accepts three numbers and calculates their average.

formula : $\text{avg} = (a + b + c) / 3$

```
In [53]: n1=int(input('Enter the value for num1:'))
n2=int(input('Enter the value for num2:'))
n3=int(input('Enter the value for num3:'))
avg = (n1+n2+n3) / 3
avg
print('value of n1 is : {} ,value of n2 is : {} , the value of n3 is: {} ,the av
print(f' n1 : {n1} , n2 : {n2} , n3 : {n3} ,the avg is : {avg}')
```

value of n1 is : 10 ,value of n2 is : 20 , the value of n3 is: 30 ,the avg is : 20.0
n1 : 10 , n2 : 20 , n3 : 30 ,the avg is : 20.0

In []: